



Sébastien MARTINET

10/07/1996
French

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Languages

English	● ● ● ●
French (native)	● ● ● ●
Spanish	● ● ● ○
Italian	● ● ● ○

Numerical tools

Python	● ● ● ●
Fortran	● ● ● ●
C/C++	● ● ● ○
R	● ● ● ○
hTTL5 & CSS	● ● ● ○

Machine Learning & Data Processing

Pandas	● ● ● ●
SQL	● ● ● ○
scikit-learn	● ● ● ○
PyTorch	● ● ● ○
TensorFlow	● ○ ○ ○
PySpark	● ● ● ○

Education

- 2022 PhD in Astrophysics**
University of Geneva,
Switzerland
- 2018 Master in Astrophysics**
Université Grenoble-Alpes,
France
- 2016 Bachelor Fundamental
Physics**
Université Pierre et Marie
Curie, Paris VI, France

Experience

- Oct. 2022 – Present. **PostDoctoral Researcher in Nuclear
Astrophysics** [Université Libre de Bruxelles, BELGIUM](#)
- Sept. 2022 – Oct. 2022. **PostDoctoral Researcher in Stellar
Astrophysics** [University of Geneva, SWITZERLAND](#)
- Sept. 2018 – Aug. 2022. **Doctoral Researcher in Stellar
Astrophysics** [University of Geneva, SWITZERLAND](#)

Data Science Projects

Statistical Modeling & Scalable Data Pipelines [Python](#), [Fortran](#),
[HPC](#), [Git](#) (7 publications - 3 first-author)

- Built a scalable pipeline to run and analyze large-scale simulations (10k+ runs) using legacy Fortran code wrapped in Python; automated deployment on HPC clusters.
- Developed and applied a statistical framework to quantify and propagate uncertainties, enabling robust predictions from high-dimensional data.
- Delivered reproducible workflows and end-to-end tooling to extract actionable insights from complex simulation output.
- Collaborated with multidisciplinary teams to translate technical results into usable observables across domains.

Enhancing Legacy Software with Modern Data Science [Python](#),
[Fortran](#), [Data Viz](#), [CLI Tools](#) (6 publications - 2 first-author)

- Led the design and integration of new physics modules into a legacy simulation code, improving performance and expanding capabilities for broader use cases.
- Modernized the data analysis interface with Python tooling (NumPy, pandas, seaborn), improving maintainability and enabling rapid visual exploration.
- Developed robust CLI tools and notebooks for data wrangling and visualization, enhancing reproducibility and team collaboration.
- Demonstrated autonomy and leadership in driving complex projects from concept to publication.

Other projects (55 collaborators, 19 institutes, 11 countries) include: integrating models with large observational datasets, analyzing experimental data to improve simulations, publishing data products, and building scientific data platforms.

Selected Publications

Martinet, S. et al. (2025) Statistical framework for nuclear parameter uncertainties in nucleosynthesis modeling of r- and i-process.
doi: [2025EPJA...61...48M](#)

Martinet, S. et al. (2024) The intermediate neutron capture process: IV. Impact of nuclear model and parameter uncertainties.
doi: [2024A&6A...684A...8M](#)

[Full list of publications](#) / [Full list of conferences](#)
(22 papers incl. 9 first author) / (15 talks in 10 different countries)